

- Q.24 What is blue tooth? Write its application. (CO4)
- Q.25 Explain concept of adjacent channel interference. (CO2)
- Q.26 What is the role of antennas in mobile communication?
Discuss the types of antennas used. (CO1)
- Q.27 What is MIMO Technology, and how does it enhance wireless communication? (CO3)
- Q.28 Explain about cell splitting. (CO1)
- Q.29 What are the factors affecting small scale fading? (CO3)
- Q.30 Explain in brief the shadowing margin. (CO3)
- Q.31 Give the features of OFDM. (CO4)
- Q.32 What is the importance of link budget? (CO2)
- Q.33 Describe in brief the feedback diversity. (CO3)
- Q.34 Describe the concept of free space path loss. (CO2)
- Q.35 Define and differentiate between macro cells, micro cells, and pico cells in mobile networks. (CO1)

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Describe the architecture and working of 5G technology, including its key features and advantages. (CO1)
- Q.37 Compare 2G/3G/4G/5G system of wireless communication.
- Q.38 Explain key components and functions of LTE technology, including its advantages in mobile communication system. (CO4)

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Roll No.

3rd Semester/ Mechtronics Subject: Mobile & Wireless Communication

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is a major advantage of cellular networks? (CO1)
- High power consumption
 - Frequency reuse
 - Large cell sizes
 - No handoff required
- Q.2 What is the process of transferring a mobile user from one cell to another called? (CO1)
- Roaming
 - Handoff
 - Frequency reuse
 - Paging
- Q.3 Which multiple access technique is used in GSM? (CO2)
- FDMA
 - CDMA
 - TDMA
 - OFDMA
- Q.4 The frequency range of 4G LTE technology is approximately: (CO4)
- 2-8 GHz
 - 700 MHz-2.6 GHz
 - 12 GHz-24 GHz
 - 60 GHz-70GHz

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- Q.5 What is the typical data rate provided by 5G technology? (CO4)
 a) 1 Gbps b) 10 Gbps
 c) 100 Mbps d) 1 Mbps
- Q.6 Which type of antennas are commonly used in mobile communication? (CO3)
 a) Dipole antennas b) Parabolic antennas
 c) Helical antennas d) Patch antenna
- Q.7 What is a pico cell in mobile networks? (CO1)
 a) A small area cell in a cellular network
 b) A large area cell
 c) A cell used for satellite communication
 d) A redundant cell tower
- Q.8 What is the percentage of voice and control channels?
 a) 95% and 5% b) 5% and 95%
 c) 50% and 50% d) 25% and 75%
- Q.9 A link budget is accounting of all _____ (CO3)
 a) Gain and losses from transmitter
 b) Power transmitted by transmitter
 c) Power received by receiver
 d) Power transmitted but not received
- Q.10 All users in CDMA uses _____ carrier frequency. (CO4)
 a) Different b) Two
 c) Ten d) Same

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SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 The process of converting analog signals into digital signals is called _____. (CO1)
- Q.12 _____ is a modulation technique used in 4G LTE. (CO4)
- Q.13 Which technique is used for channel access GSM? (CO3)
- Q.14 Wi-Fi operates mainly in the _____ GHz frequency band. (CO2)
- Q.15 The process of reducing interference in wireless communication is called _____. (CO3)
- Q.16 Define path loss. (CO2)
- Q.17 The basic unit of a cellular system is called a _____. (CO2)
- Q.18 Bluetooth operates in the _____ GHz band. (CO1)
- Q.19 Define Fading. (CO2)
- Q.20 Define Cell sectoring. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the GSM architecture. (CO1)
- Q.22 What is a hand off in mobile communication and what are the different types of hand off? (CO1)
- Q.23 How does CDMA differ from TDMA and FDMA in terms of multiple access techniques? (CO4)

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